

Requirements Document

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1. Introduction

This Requirements Document defines the functional and non-functional requirements for the KSUMB Attendance System. It describes what the system must do and the quality attributes it must exhibit, without prescribing a specific technical implementation. All requirements trace back to the business objectives established in the Project Charter and the stakeholder needs identified during the interview with Sharyn Worcester and Dr. Wimmer.

This document serves as the foundation for downstream planning artifacts including the Scope Statement, Work Breakdown Structure, Schedule, and ultimately the Design and Test Plans.

2. Stakeholders & User Roles

Role	Description	Primary System Capabilities
Staff/Director	Band directors and administrative staff (e.g., Dr. Wimmer, Sharyn Worcester). Full system authority.	- Real-time dashboard - global roster view - per-student history - exception approval - location/threshold configuration - reporting & export - role management - kiosk administration.
Section Leader	Student leaders responsible for a specific section. Elevated permissions scoped to their section.	- Dynamic section roster with photos - manual check-in of members - set section pre-check-in location/time - view exception info (configurable by staff) - self-check-in.
Member	General band members (~400+ students). Standard permissions.	- Self-check-in (GPS-verified) - Submit Pink Sheets and Yellow Sheets - view personal attendance history.
Kiosk User	Any member who cannot use their personal device. Authenticated via student ID scan or manual login.	Self-check-in only. No SSO required; standalone fallback authentication.

Note: A staff-configurable setting shall control whether section leaders can view the reason/details for an approved exception. For example, a head section leader may see reasons while other section leaders may not.

3. Functional Requirements

Note: Requirements are prioritized using MoSCoW: Must (M), Should (S), Could (C), Won't this phase (W).

3.1 Attendance Check-In

ID	Requirement	Priority	Acceptance Criteria
FR-101	The system shall allow members to self-check-in via a mobile web interface by confirming their GPS location is within a defined geofence.	S	- A member within the geofence radius can successfully check in - a member outside receives an error.
FR-102	Staff shall be able to define check-in locations from a set of pre-defined options (e.g., Band Hall, Memorial Stadium, Brandeberry Complex) and also create custom locations by dropping a pin on an interactive map.	S	Staff can select a pre-defined location or drop a pin; the geofence center updates accordingly and is visible to all users.
FR-103	Custom check-in locations can be set during event creation or adjusted on the fly before/during an event.	M	A staff member can modify the check-in location of an active event, and the change propagates to all users in real time.
FR-104	Section leaders shall be able to independently set a "section pre-check-in" location and time for their section (e.g., a game-day meetup point before the official call time).	C	Section leaders can set location/time without staff approval; section members see the pre-check-in info in their view.
FR-105	Section leaders shall be able to manually check in members who forgot their phone.	M	A section leader can select a member from their roster and record a check-in on their behalf with a timestamp.
FR-106	The system shall provide a kiosk mode where students can check in by scanning their K-State student ID or by manual login (username/ID number). The kiosk operates as a standalone fallback and does not require SSO.	S	A student at the kiosk can scan their ID card or type credentials and receive a check-in confirmation.
FR-107	Each check-in shall record: student ID, timestamp, GPS coordinates (or kiosk/manual indicator), and the event it is associated with.	M	All check-in records contain these four data fields and are queryable by staff.
FR-108	The system shall support offline check-in caching: if the device has poor or no connectivity, the check-in request is stored locally and submitted automatically when connectivity is restored, preserving the original timestamp and location.	C	A check-in made offline appears in the system with its original timestamp once connectivity resumes. Cached check-ins are visually flagged for staff review.
FR-109	The system shall allow staff members to configure the radius and location of the geofencing areas per event.	M	Staff can choose from a set of coming locations or create a custom location for required events

3.2 Tardy Thresholds & Logic

ID	Requirement	Priority	Acceptance Criteria
FR-201	The system shall calculate tardy status based on configurable, role-based thresholds. Default values: Section Leaders = 10 minutes, Members = 20 minutes after call time.	C	A member arriving 19 minutes after call time is on time; at 21 minutes they are tardy. Thresholds are editable by staff.
FR-202	Exception modifiers shall be additive to the base threshold. A student with a class conflict (Yellow Sheet) receives an additional +30-minute grace period on top of their role-based threshold.	S	A member with an approved class conflict arriving 49 minutes late is on time (20 + 30 = 50 min threshold).
FR-203	Staff shall be able to configure all tardy threshold values globally and override them on a per-event basis.	M	Staff can set global defaults and modify them for a specific event; event-level overrides do not affect other events.

ID	Requirement	Priority	Acceptance Criteria
FR-204	Tardy thresholds and exception modifiers shall be configurable per role type and per exception type from an admin settings panel.	C	Staff can add or modify threshold rules (e.g., drum majors get 5 minutes, color guard gets 15 minutes) without developer intervention.

3.3 Exception Management (Pink & Yellow Sheets)

ID	Requirement	Priority	Acceptance Criteria
FR-301	The system shall allow members to submit a Pink Sheet (one-off absence request) for any upcoming event. If the submission is within 24 hours of the event, the system displays a warning to the student and formats the "time submitted" field differently (e.g., highlight/color) in the staff view.	M	A Pink Sheet submitted 23 hours before the event shows a visual warning to the student and appears highlighted in the staff queue.
FR-302	Students may submit a Pink Sheet after the event has already occurred. Late submissions are flagged for staff and require explicit staff approval.	C	A post-event Pink Sheet is clearly marked as "Late Submission" in the staff view and defaults to pending until staff acts.
FR-303	Staff shall approve, deny, or request more information on Pink Sheet submissions. Approved Pink Sheets result in the attendance record showing as "Excused".	M	An approved Pink Sheet changes the student's status for that event from "Absent" to "Excused" in all views and reports.
FR-304	The system shall allow members to submit a Yellow Sheet (recurring class conflict) once at the beginning of the semester. The Yellow Sheet captures: class name, instructor, class time, days of the week, and any additional notes.	M	A student submits a Yellow Sheet with all required fields; the form rejects incomplete submissions.
FR-305	Staff shall verify and approve Yellow Sheets. Once approved, the system automatically applies the class conflict exception modifier to all applicable events for the semester.	M	After approval, the student's tardy threshold is automatically extended for every event that overlaps with the documented class time.
FR-306	Staff shall have the ability to revoke or modify a previously approved Yellow Sheet at any time during the semester.	S	Revoking a Yellow Sheet removes the exception modifier from future events; past records remain unchanged.

3.4 Dashboards & Views

ID	Requirement	Priority	Acceptance Criteria
FR-401	The Staff Dashboard shall provide a real-time, high-level overview of attendance across all sections for the current event, including total present, absent, tardy, and excused counts.	M	Staff can view a live summary that updates as check-ins occur, with per-section breakdowns.
FR-402	Staff shall be able to drill down into any section to see individual member status, and into any member to see their full attendance history for the semester.	M	Clicking a section shows its roster with statuses; clicking a member shows a chronological history of events.
FR-403	The Section Leader view shall display a dynamic roster of their section members with photos, current check-in status, and relevant exception information.	M	Section leaders see an auto-updating roster with member photos and status indicators (present, absent, excused, tardy).
FR-404	Whether section leaders can see the reason for an approved exception shall be controlled by a staff-configurable, per-leader setting.	C	Staff can toggle exception-reason visibility independently for each section leader from an admin panel.
FR-405	The Member view shall show the student's own check-in interface and a personal attendance history (events attended, tardy, absent, excused).	M	Members can see their history but cannot see other members' records or any grade-related information.

ID	Requirement	Priority	Acceptance Criteria
FR-406	Student grades shall not be visible in the system. The system provides raw attendance and exception data only; grade determination is performed by staff outside the system at year-end.	M	No grade calculations or grade fields exist in any user-facing view.
FR-407	Staff shall be able to generate and export attendance reports (e.g., CSV/Excel) for use in external grading workflows.	S	Staff can filter by date range, section, or student and export a downloadable file.
FR-408	The system shall generate an automated Daily Consolidated Concern Report flagging unexcused absences and problematic tardies, delivered to staff (e.g., email at a configured time such as 5:30 PM).	C	Staff receives a summary report after each event day without manual action.

3.5 Event & Roster Management

ID	Requirement	Priority	Acceptance Criteria
FR-501	Staff shall be able to create, edit, and delete events. Each event includes: name, date, call time, check-in location, and applicable tardy threshold overrides.	M	Staff can create an event with all required fields; events appear in all user views.
FR-502	The system shall support bulk import of roster data from the existing systems.	M	Staff can upload a FileMaker export file and the system parses and imports student records.
FR-503	Staff shall have tools to manually edit user information, roles, section assignments, and attendance records.	M	Staff can reassign a student to a different section, change a role, or correct a check-in record.
FR-504	The system shall support multiple course numbers (100/400/800 level) for students enrolled in different band credit sections.	S	Students are tagged with their course number; reports can be filtered by course level.
FR-505	The system shall maintain an immutable audit log of all check-in and status change events	C	Audit log is exportable through a CSV

3.6 Notifications / Communications

ID	Requirement	Priority	Acceptance Criteria
FR-601	Students get notified when their Pink/Yellow sheet is approved/denied	W	Staff can create an event with all required fields; events appear in all user views.
FR-602	Staff gets notified when students submit a Pink/Yellow sheet for approval	C	Staff receives an email or SMS when pending requests exist
FR-603	The system will notify maintainers when there are server issues or excessive errors.	W	Performance monitoring dashboard with automatic alert thresholds

4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Priority	Acceptance Criteria
NFR-01	The system shall handle 400+ simultaneous check-in requests at approximately 3:00–3:30 PM without degradation.	S	Load testing demonstrates 400 concurrent check-ins complete within 3 seconds each.
NFR-02	The web interface shall be lightweight with minimal asset sizes to perform acceptably on variable outdoor cellular/Wi-Fi connections.	S	Initial page load is under 500 KB; check-in action completes in under 2 seconds on a 3G connection.

4.2 Reliability & Availability

ID	Requirement	Priority	Acceptance Criteria
NFR-03	The system shall implement offline check-in caching so that check-ins are not lost due to intermittent connectivity.	C	A cached check-in is queued locally and synced when connection resumes, preserving original timestamp and GPS data. Cached entries are visually flagged for staff.
NFR-04	The system shall include graceful error handling: informative error messages for failed actions and retry logic for network requests.	M	Users see a clear message on failure; the system retries failed requests up to 3 times before showing an error.

4.3 Security & Compliance

ID	Requirement	Priority	Acceptance Criteria
NFR-05	The system shall use K-State University Single Sign-On (SSO) / Active Directory for primary authentication of all staff, section leaders, and members.	S	Users log in with their K-State eID; no separate credential store is required for standard logins.
NFR-06	The kiosk mode shall operate with standalone fallback authentication (student ID scan or manual entry) and shall not require SSO.	S	A kiosk check-in is successfully recorded using a student ID scan without an SSO session.
NFR-07	The system shall comply with FERPA guidelines and K-State data privacy policies. Student attendance records shall be treated as educational records.	M	No student data is exposed to unauthorized users; role-based access is enforced at the API level.
NFR-08	Role-based access control shall enforce strict data boundaries: section leaders see only their section; members see only their own data.	M	A section leader cannot access roster data or attendance records for sections other than their own.

4.4 Usability

ID	Requirement	Priority	Acceptance Criteria
NFR-09	The interface shall be mobile-first and responsive, optimized for smartphone screens but functional on tablets and desktops.	S	All core workflows (check-in, exception submission, section roster) are usable on a 375px-wide screen.
NFR-10	The UI shall follow a minimal-click design philosophy: the most common action (self-check-in) should require no more than 2 taps from the home screen.	C	User testing confirms check-in is achievable in 2 taps or fewer.
NFR-11	Clear color-coding or iconography shall indicate attendance statuses (present, absent, tardy, excused), backed by structured data for accessibility.	C	Statuses are distinguishable by both color and icon/label; color alone is not the only differentiator.

4.5 Hosting & Sustainability

ID	Requirement	Priority	Acceptance Criteria
NFR-12	The system shall be hosted and owned by Kansas State University (e.g., K-State IT managed hosting or a university service account such as ksumb@ksu.edu).	C	The production environment is under a K-State institutional account; no individual student accounts are required for operation.
NFR-13	The system shall not rely on any single student's personal account for maintenance or operation. It must be transferable to K-State IT or future band staff.	M	Documentation and access credentials are stored in institutional accounts; a handover guide is provided.
NFR-14	The solution shall avoid expensive commercial software licenses. Low-to-zero ongoing operational cost is preferred.	C	Ongoing monthly hosting costs do not exceed the band program's existing software budget.

5. Technology Preferences

While this document focuses on what, not how, the following technology preferences have been expressed by the project team and sponsor. These are recorded as constraints rather than requirements and will be formally evaluated during Design.

Constraint	Detail
Frontend	Web-based interface, likely React or Next.js.
Connectivity	Wi-Fi available at field; app should be lightweight with offline caching for poor connections.
Authentication	K-State SSO / Active Directory for primary auth; standalone fallback for kiosk.
Data Integration	Import roster data from KSIS Registry / Onboarding Process
Hosting	K-State IT managed or institutional service account. Off-campus hosting requires IT review.

6. Assumptions & Dependencies

6.1 Assumptions

- All band members have access to a smartphone with a modern web browser and GPS capability.
- K-State IT will cooperate on SSO integration and any required security reviews.
- The existing FileMaker Pro roster export format will remain stable or a mapping will be provided. At the very least, it will serve as example data for development.
- Wi-Fi coverage at Memorial Stadium and the practice field is sufficient for most users, with offline caching covering gaps.
- Off-the-shelf attendance solutions have been evaluated and do not meet KSUMB-specific workflow requirements.

- Tardy threshold values provided (10/20/+30 min) are initial defaults; staff will finalize exact values before deployment.

6.2 Dependencies

- Creation of a KSU service account for the KSUMB that will be used for hosting services.
- K-State IT approval for SSO integration and/or hosting arrangements.
- Confirmation of FERPA compliance requirements from the university's data privacy office.
- Stakeholder availability for requirement validation and user acceptance testing.

7. Out of Scope (This Phase)

- Integration with KSIS (university Registrar) for automatic roster or schedule importing.
- Pep-band sign-up system.
- Functionality for other K-State band programs (Wind Symphony, Concert Band, etc.).
- Calendar view for all events.
- Automated grade calculation or grade display within the system.

8. Open Questions & Risks

The following items require further clarification or carry risk that may affect downstream planning:

#	Item	Impact / Notes
1	Kiosk hardware procurement: What device(s) will serve as kiosks? Who provides and maintains them?	• Affects FR-106, NFR-06. May require budget allocation and IT coordination.
2	K-State IT SSO timeline: How long does the SSO integration approval and setup process take?	• Critical path risk. Delays here push the entire project.
3	Daily Concern Report delivery method: Is email sufficient, or do stakeholders want in-app notifications or Slack/GroupMe integration?	• Affects FR-408 scope. Defer to design phase if email is baseline.
4	Offline caching edge cases: What happens if a student checks in offline but never regains connectivity during the event window?	• Need a policy decision: auto-expire cached check-ins after X hours, or always honor them?
5	Hosting: What service should be used to host the web service and database?	• Many different options are available; will depend on KSU IT approval, price, availability, and ease of use • Most promising current options: Google Firebase, Loveable, Vercel, Supabase...